

Introduction to Python Programming

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Concept of Programming

► What is Programming?

The process of creating a set of instructions that tell a computer how to perform a task.

► Why Learn Programming?

- Automate tasks.
- Solve problems efficiently.
- Develop software and applications.

► Key Components:

- Algorithms.
- Writing, testing, and debugging code.

Define a Computer Program

- ▶ What is a Computer Program?
 - A sequence of instructions written to perform a specific task.
- ▶ How it Works:
 - Input: Data provided by the user or environment.
 - Process: Logical steps executed by the program.
 - Output: Result of the processed input.
- ▶ Example:
 - A calculator program to perform arithmetic operations.

Concept of Low level and High Level Language

- ▶ A **low-level language** is a programming language that is close to machine code or hardware, offering minimal abstraction and direct control over system resources.
- ▶ A **high-level programming language (HLL)** is a programming language that is designed to be easy for humans to read, write, and understand.
- ▶ Examples :Python,Java,C#,JavaScript etc

What is Python Programming?

► Definition:

- Python is a high-level, interpreted programming language known for its simplicity and versatility.

► Key Features:

- Easy to learn and use.
- Extensive standard library.
- High level Language

► Uses of Python:

- Web development, data analysis, machine learning, automation, and more.

Basics of Python

► Getting Started:

- Install Python: python.org
- Use an IDE: VS Code, PyCharm, or Jupyter Notebook.

► First Python Program:

```
print("Welcome to the World of Python!")
```

Variables in Python

- ▶ A variable in Python is a symbolic name that refers to a memory location where data is stored. It acts as a container to hold values that can be used and manipulated during program execution.
- ▶ Keywords are reserved words in python
- ▶ Identifiers are Class/Function/Variable names

Key Points About Variables

- ▶ **Dynamic Typing:** Python allows variables to hold values of any data type, and the type can change during runtime.
- ▶ **No Explicit Declaration:** Variables do not require explicit declaration or data type specification. Python automatically assigns the type based on the value.

Rules for Defining Variables in Python

- ▶ Variable names must begin with a letter (a-z, A-Z) or an underscore (_).
- ▶ **No Special Characters:**
Variable names can only contain letters, numbers, and underscores (_).
- ▶ Cannot be a Python keyword
- ▶ Case Sensitivity
- ▶ **No Spaces:**
 - ▶ Variable names cannot contain spaces

Data Types in Python

- ▶ **Data types** define the kind of data a variable can store in a programming language, such as numbers, text, or boolean values.
- ▶ **Primitive Data types:**
 - ▶ Integer
 - ▶ Float
 - ▶ String
 - ▶ Boolean
 - ▶ None
- ▶ **Derived Data types:**
 - ▶ List/Array
 - ▶ Tuple
 - ▶ Set
 - ▶ Dictionary